

ESS IA 2026:

Investigating the Impact of Concrete on Soil Fertility (A focus on pH and nutrient composition)

Research Question

To what extent does the proximity to concrete affect the nutrient composition of soil (Nitrogen, Phosphorus) and its pH in Boston, Massachusetts a focus on the Chestnut Hill Reservation

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Criterion A: Research question and inquiry

Introduction / Background of Global Context

Humanity, from its existence has constantly sought to urbanize the world which has emerged as a pressing concern for policymakers and scientists globally. Due to the natural increase of the population, greater amounts of immigration, and the longevity of people increasing, urbanization has drastically increased. Also, due to the increased opportunities in cities, rural to urban migration has also increased. Urbanization clears and destroys pre-existing ecosystems and habitats and since 1990, around 420 million hectares of forest have been lost due to our colonization of the natural world. However, the effects of urbanization aren't just short-term, and heavy metals, petroleum hydrocarbons, road salts and other nutrients seep into surrounding ecosystems (U.S. Environmental Protection Agency, 2003).

Since 1950, global cement production has increased more than 30-fold, and this dramatic increase in concrete has direct effects on ecological habitats and particularly the soil (Andrew, 2019). This cement is used to build buildings which are an essential part of urbanization. Global urbanization rates in 2024 were 57.9% and within the United States it was 83% illustrating that the majority of the population is living in urban areas (Ritchie et al., 2024). Concrete is a heavy and non-porous substance that compresses soil reducing pore space and preventing aeration, limiting water infiltration (Yang & Zhang, 2011). Thus, in places of high urbanization, there would be high uses of concrete. This impermeableness acts as a barrier to the water cycle and can lead to dryness or water logging in certain areas. Also, concrete contains compounds like calcium hydroxide, sodium and other ions that directly alter the chemical composition of the soil. Therefore, as urbanization and concrete are closely linked, I aimed to understand the environmental impacts of urbanization which mostly manifests as an increase in concrete use.

Introduction in Local Context

Figure 1: Map of cover of Location of Boston (Fischman, n.d.)



Figure 2: Map of Land Cover of Boston Area (National Research Council, 2013).



Boston is a metropolitan city in the state of Massachusetts located in the Northeast region, within the United States (see figure 1). Boston is a biome consisting mainly of a temperature deciduous forest. It is considered a highly urban city, and houses 9.5% of total Massachusetts population (US Census Bureau, 2021). Increased levels of urbanization began in 1961, and since then, Boston's usage of concrete has grown significantly. It is estimated that 27% of Boston is covered by impervious substances, including concrete and asphalt (see figure 2).

However, despite the minimal area covered by concrete, the quality of soil surrounding concrete hasn't been extensively studied. Therefore, the relationship between the local and global issues highlights the need for a comprehensive study of the extend of proximity to concrete impacting soil quality. This research area directly links to topic 8.2 being *Urban systems and urban planning* and 5.1 being *Soil*.

Specifically close to where I live in Boston, there exists the Chestnut Hill Reservation. This is a park densely populated with trees and other wildlife and surrounded by concrete roads. There are some small hiking trails in the park, but it is primarily a nature reservation. Therefore, this study used the Chestnut Hill Reservation for soil sampling which follows their rules.

Research Question

To what extent does the proximity to concrete affect the nutrient composition of soil (Nitrogen, Phosphorus) and its pH in Boston, Massachusetts in the Chestnut Hill Reservation

This question both addresses a larger global issue, as a recent study highlighted that 95% of Earth's fertile soil will be degraded by 2050 mainly due to unsustainable farming (Soil S, 2024). Specifically, this links to *ESS 5.1 topic*, where the integrity of soil is studied. Other studies highlight how 75 billion tons of soil is annually globally lost (Soil Degradation: The Silent Global Crisis, n.d.). However there have been no global or local studies on the impact of humanities growing demand of cement on soil quality. Globally, 30 billion metric tons of concrete are produced each year, making it the most consumed material on Earth after water

(Xiao et al., 2025). Therefore, this study aimed at understanding the local impacts of concrete, as it will begin to play an increasing role in our lives in the future. Furthermore, this links to the *ESS topic 8.2*, which discusses urbanization and suburbanization and how this directly influences the ecosystems and systems.

Criterion B: Strategy

Within Boston, there are preexisting strategies the government monitors soil health. The Massachusetts Department of Agricultural Resources (MDAR) provide technical assistance in the form of funding to programs designed at collecting data and creating a comprehensive database for Massachusetts (Commonwealth of Massachusetts, n.d.). MDAR not only offers grants for landowners to promote sustainable soil practices, but they promote composting and nutrient management to ensure soil fertility. Also, other organizations like the Executive Office of Energy and Environmental Affairs (EEA) utilize the soil carbon content as an indicator of the overall health of the soil and engage stakeholders to monitor the effectiveness of numerous conservation practices. While there exist programs and organizations that mitigate and monitor soil health, none specifically monitor nutrient or pH and absolutely none track the effects of concrete on soil health.

Social, Economic, Political, Environmental and Cultural Tensions

Table 1: Summarizing the different tensions in strategies implemented

Social and Economic	East Boston is a lower income community where there have been repeated protests due to the increasing costs that could be incurred due to climate change policy (Extinction Rebellion Boston, n.d.), thus possessing an Anthropocentric perspective. Because there exist different soil management systems, when implemented in an urbane setting, marginalized communities often get displaced. In Boston, there existed a variety of tensions between different economic groups as these different groups of people would receive different amounts of assistance, thus creating tensions between people with Anthropocentric and Ecocentric perspectives.
Political	Different political parties may oppose government intervention of private land. For example, Michele Wu, the Mayor of Boston, championed the ‘Green New Deal for Boston’, which address climate change, pollution and equitable access to green spaces (Michelle for Boston, n.d.). All of these link to adequate management of soil systems. However, the previous administration wasn’t keen in investing in sustainable development and thus created political tensions. This highlights the tensions that are created by Technocentric and Ecocentric individuals.
Cultural	Indigenous groups, like the Wampanoag or the Massachusett people in Boston, typically have an eccentric and stewardship perspective as they typically understand the direct impacts on their land. However, employees in companies that gain direct benefit from pollution of our ecosystems like oil or gas companies in Boston may possess an anthropogenic perspective

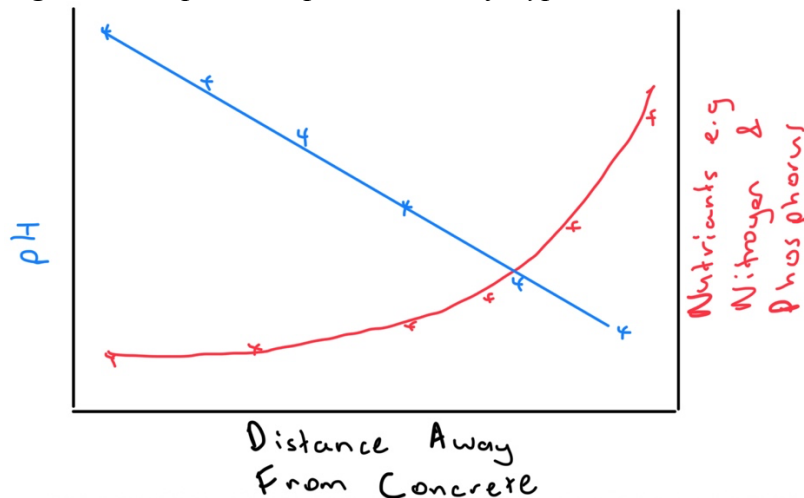
	thus creating tensions. Also, other parts of Boston society may possess a Technocentric perspective and advocate for new technologies to monitor and improve soil fertility (Jburke, 2024).
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Conflicting Goals of Stakeholders

Land and the soil are occupied by private and public landowners. Approximately 36% of land in Massachusetts is publicly owned with the remaining 64% privately owned (Palmieri et al., 2011). While typically, private landowners seek to maximize yield of their soil, they seek minimal intervention from the government whereas public sector requires soil conservation techniques introducing conflicting goals. Also, environmentalists with an eccentric perspective seek to monitor and track pH and nutrients to ensure a healthy soil system, whereas other local and indigenous communities push for economic gains to maximize output of their land. Therefore, balancing the socioeconomic, political and cultural tensions with the goals of stakeholders is crucial to ensure the policy is equitable.

Hypothesis

Figure 3: Graphical Representation of Hypothesis



My hypothesis states that as proximity to concrete decreases, the pH should drop and other nutrients like Nitrogen or Phosphorus would rise again (see figure 3). Because concrete contains calcium compounds like calcium hydroxide (which are alkaline) the closer the soil is to concrete, the greater the pH (Li et al., 2020). This rise in pH affects nitrogen and phosphorus cycling in the soil and affects the microorganisms in the soil. Thus, it was hypothesized that concrete reduces both these nutrients and its impact on such nutrients reduces the farther away the soil is sampled from the concrete (Mao et al., 2023).

Variables

Independent Variable	Units	Range	Method to Manipulate
Distance from concrete source	meters	0, 20, 40, 60, 80, 100	Exact distance was determined using a digital tape measure
		The distances were selected to accommodate for the size of the area that was sampled. Also, intervals of 20 meters were chosen to ensure that accurate relationship could be determined.	
			Uncertainty of balance = ± 0.05 meters

Dependent Variable	Units
The pH, Nitrogen and Phosphorus levels	pH is reported on a logarithmic scale from 1-14 Nitrogen and Phosphorus are estimated in parts per million (ppm)
	A Nitrogen, Phosphorous and pH soil test kit was used to determine the nutrient levels

Control Variables	How it was controlled	Impact on Outcome
Soil type	The soil type was tested in the preliminary experiments, and was consistent throughout the experiment	If different soil types were used, each has different productivity and thus would have different nutrient cycling which would change the Nitrogen, phosphorus and the pH of the soil
Temperature and light exposure	All soil samples were extracted between 1:00 – 3:00 pm to ensure that the temperature and light was constant	If samples were taken when temperature or light was variable, this could directly affect the nutrients in the soil
Duration between testing	All soil was tested 30 minutes after it was collected	If too much time elapses after the soil is collected, nitrogen levels can dramatically change due to the changes in microbial activity (Agronomy eUpdate September 8th, 2022 : Issue 923, n.d.).
Amount of soil collected	100-gram soil sample was extracted for each trial	Naturally if more soil was used, there would be a greater amount of nutrients, which would skew the results

Confounding variable	Impact on Reaction
Plant species	Different plant species like legumes might be involved in nutrient cycling at different rates, and thus directly affects the nutrient composition of the soil
Watering schedule	Different watering might wash away nitrogen and phosphorus, and dilute the hydrogen ions thus changing the pH
Soil microbial activity	The microbial activity in the soil directly relates to the nitrogen and phosphorus as microbes are involved in decomposition of organisms producing such nutrients (Wang et al., 2024).

Criterion C: Method

Apparatus

Apparatus	Uncertainty
Soil Test Kits	± 5 ppm
Digital Tape Measure	± 0.5 m
Measuring cylinder (100 mL)	± 1 mL
Digital balance	± 0.01 g
Soil Test Color Chart	
Shovel	
6 Sealable Clear Bags	
Compass	

Methodology

Figure 4: Soil Sampling Site Distribution Map of Chestnut Hill Reservation



Obtaining soil samples

1. Set up a transect of 100 meters starting at Location A from the Chestnut Hill Reservation (see figure 4)
2. Use a shovel and dig 5 holes, each roughly 5 centimeters deep (see figure 5) starting at Location A, 0 meters from concrete

Figure 5: Image of 4 Holes Dug



- a. Use a mass balance and extract 100 grams of soil from each hole and place it in a clear bag and seal it. Then keep it in a collection box
3. Repeat steps 1-2, collecting soil samples at the new location 20 meters from the previous (see figure 4) and moving further down ensuring that all samples are collected on the transect
 4. Any residual soil from testing was returned to its original location

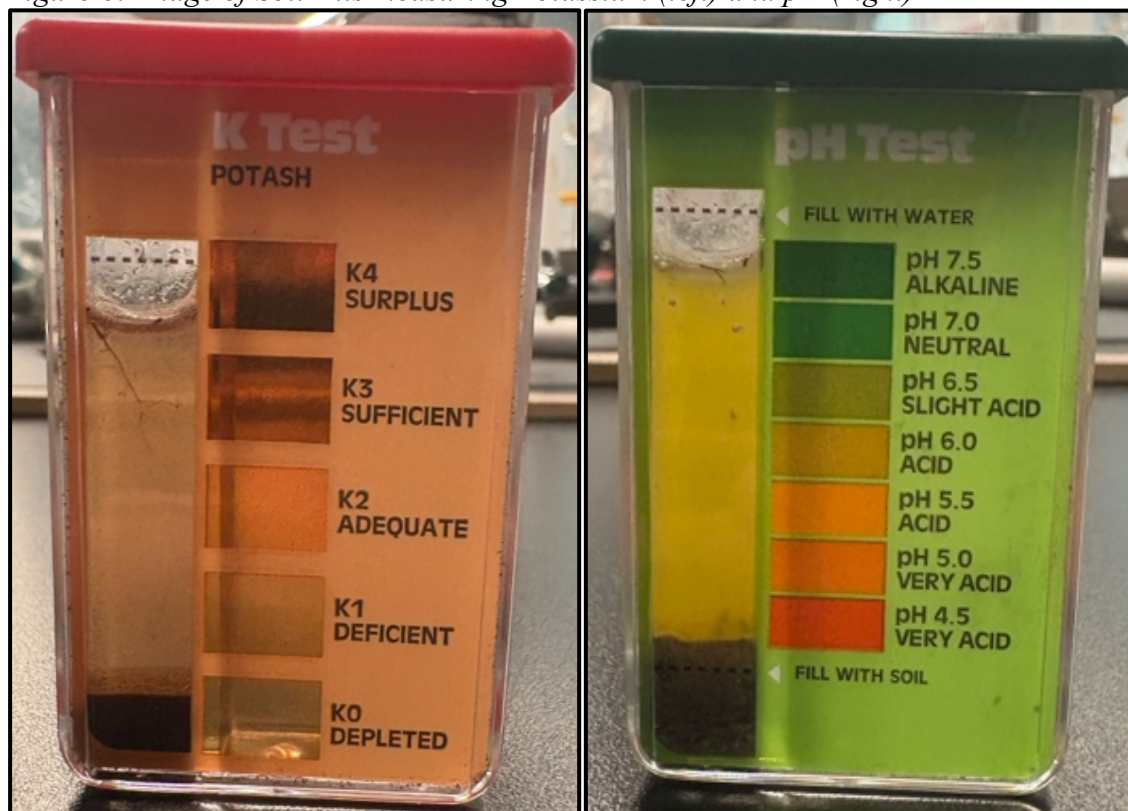
Testing Soil

1. Measure 5g sample of soil and put it in the soil testing kit (see figure 5 below)
2. Add water till the mark on the soil kit and empty a Nitrogen capsule in the kit
 - a. Repeat using the Phosphorous and pH capsule
3. Wait 10 minutes
4. Compare the color to the color sheet and estimate the concentration of nutrient
5. Note down concentrations in this table:

Data Collection Table 1:

Distance From Concrete	Nutrient Levels				
	Trial 1	Trial 2	Trial 3	Trial 4	Trial 5
0.0					
20.0					
40.0					
60.0					
80.0					
100.0					

Figure 6: Image of Soil Kits measuring Potassium (left) and pH (right)



Justification for Method

This is an accurate method for measuring the nutrients in the soil given the time frame given to conduct the Internal Assessment. Also the school restricts the budget for each project, and these kits provided the best economic value for their precision and were thus used. The soil test kits were chosen for their accuracy and precision in identifying the concentrations of nutrients. Also, this a systematic sampling strategy was used to determine the relationship between the distance from concrete and its nutrient composition. Using intervals of 20 meters allows an accurate assessment or the relationship.

Safety, Environmental or Ethical Considerations

Substance	Danger	Mitigation
Soil Kits	Safety issue – the reagents in the soil kits can be irritants and mildly toxic Environmental – improper disposal could contribute to pollution	Wear gloves and eye-protection and wear eye-protection. Waste was disposed in accordance with school's lab safety protocol
Soil	Ethical issue – habitats and ecosystems could be disturbed due to sampling Environmental – oversampling of soil could lead to permeant damage to habitats	Only sample limited soil and ensure that all soil is returned to its original location

Criterion D: Treatment of Data

Raw Data and Processed Data

Table 1: Distance from Concrete vs. pH Levels

Distance From Concrete\ m (± 0.5)	pH Levels / pH (± 0.25)					Average	Standard Deviation
	Trial 1	Trial 2	Trial 3	Trial 4	Trial 5		
0.0	10.8	9.8	10.9	10.2	9.7	10.3	0.6
20.0	9.0	8.2	6.9	7.8	6.5	7.7	1.0
40.0	6.5	7.2	7.0	7.5	7.5	7.1	0.4
60.0	6.4	7.1	7.4	7.2	6.3	6.9	0.5
80.0	5.9	6.4	6.7	6.3	6.5	6.4	0.3
100.0	6.1	5.3	5.9	5.5	5.0	5.6	0.4

Graph 1: Scatterplot with Trendline Highlighting Distance from Concrete vs. pH Levels

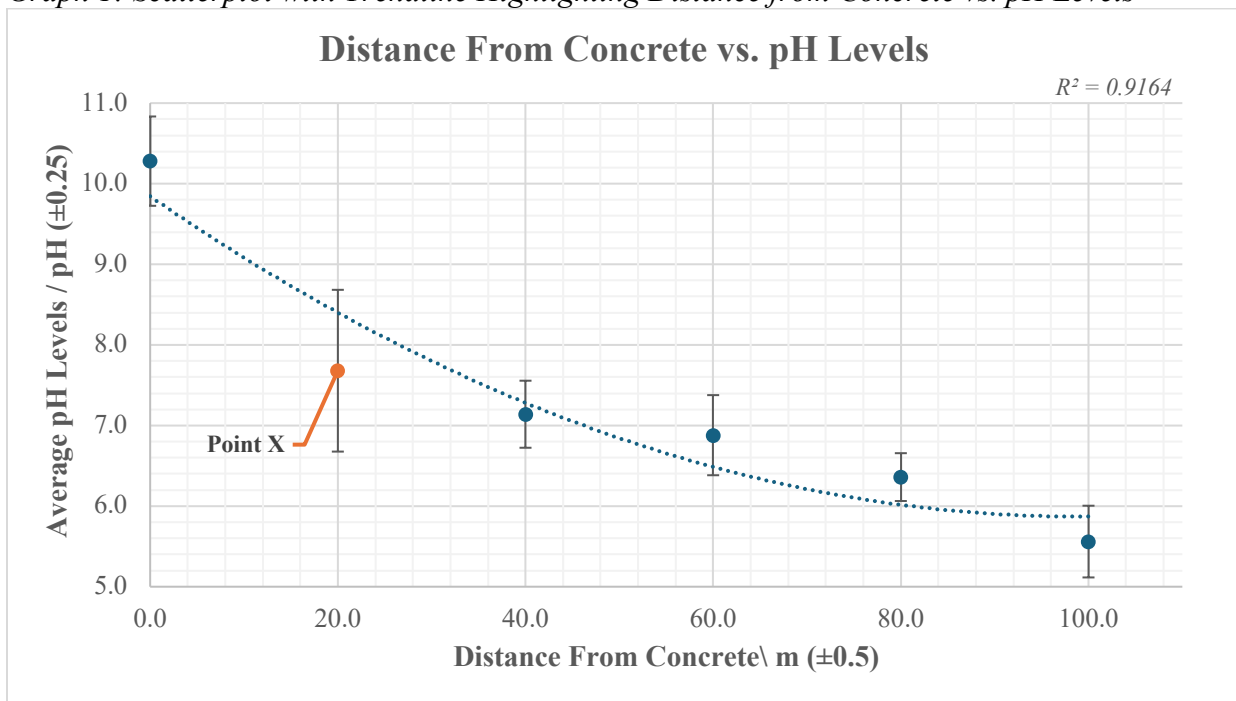


Table 2: Distance from Concrete vs. Nitrogen Levels

Distance From Concrete\ m (± 0.5)	Nitrogen Levels / ppm (± 0.5)					Average	Standard Deviation
	Trial 1	Trial 2	Trial 3	Trial 4	Trial 5		
0.0	3.1	7.5	5.1	10.6	4.2	6.1	3.0
20.0	5.6	9.1	6.8	10.5	9.8	8.4	2.1
40.0	8.3	9.6	18.6	11.3	11.4	11.8	4.0
60.0	23.5	22.5	29.1	25.4	20.4	24.2	3.3
80.0	24.8	26.3	29.1	27.7	23.5	26.3	2.2
100.0	36.4	32.8	39.0	34.1	35.6	35.6	2.4

Graph 2: Scatterplot with Trendline Highlighting Distance from Concrete vs. Nitrogen Levels

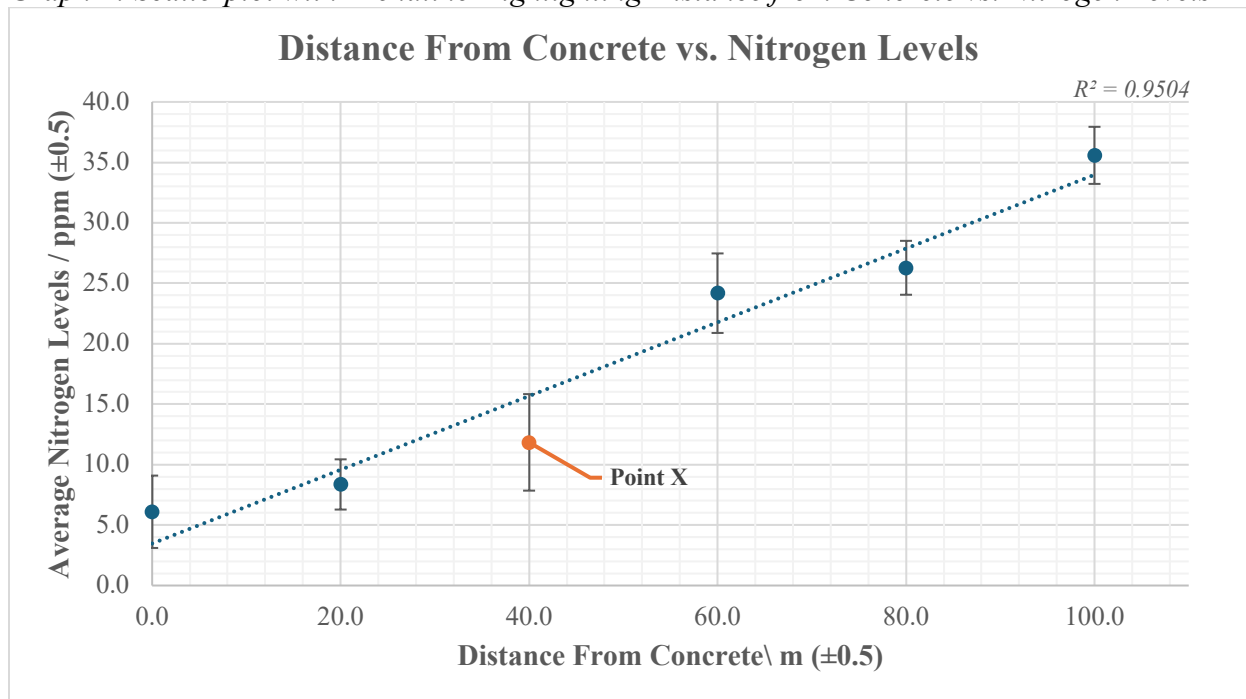
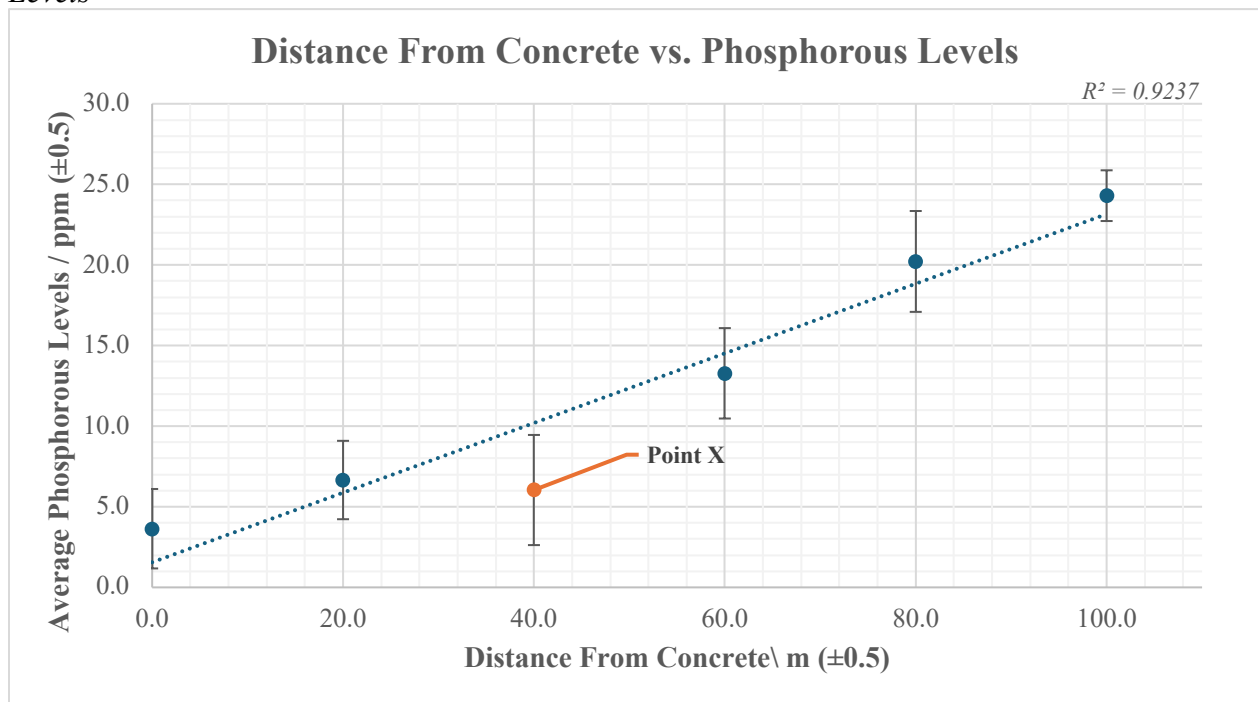


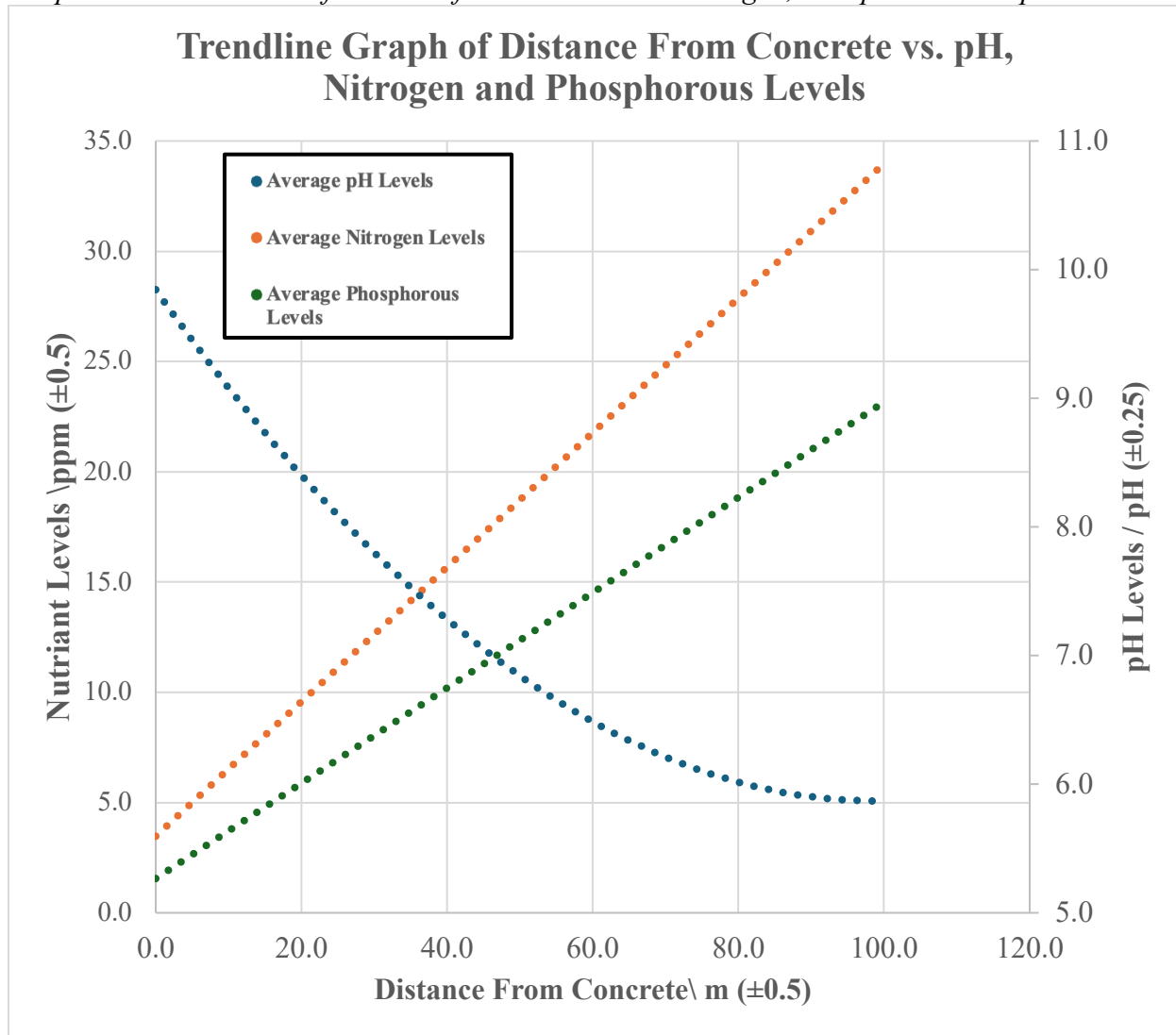
Table 3: Distance from Concrete vs. Nitrogen Levels

Distance From Concrete m (± 0.5)	Phosphorous Levels / ppm (± 0.5)					Average	Standard Deviation
	Trial 1	Trial 2	Trial 3	Trial 4	Trial 5		
0.0	2.3	1.8	4.5	7.6	2.0	3.6	2.5
20.0	9.7	4.2	5.5	8.8	5.1	6.7	2.4
40.0	5.5	9.4	3.4	2.2	9.7	6.0	3.4
60.0	17.3	13.7	9.5	13.5	12.4	13.3	2.8
80.0	24.3	16.4	19.3	22.4	18.7	20.2	3.1
100.0	26.7	22.5	24.3	24.6	23.4	24.3	1.6

Graph 3: Scatterplot with Trendline Highlighting Distance from Concrete vs. Phosphorous Levels



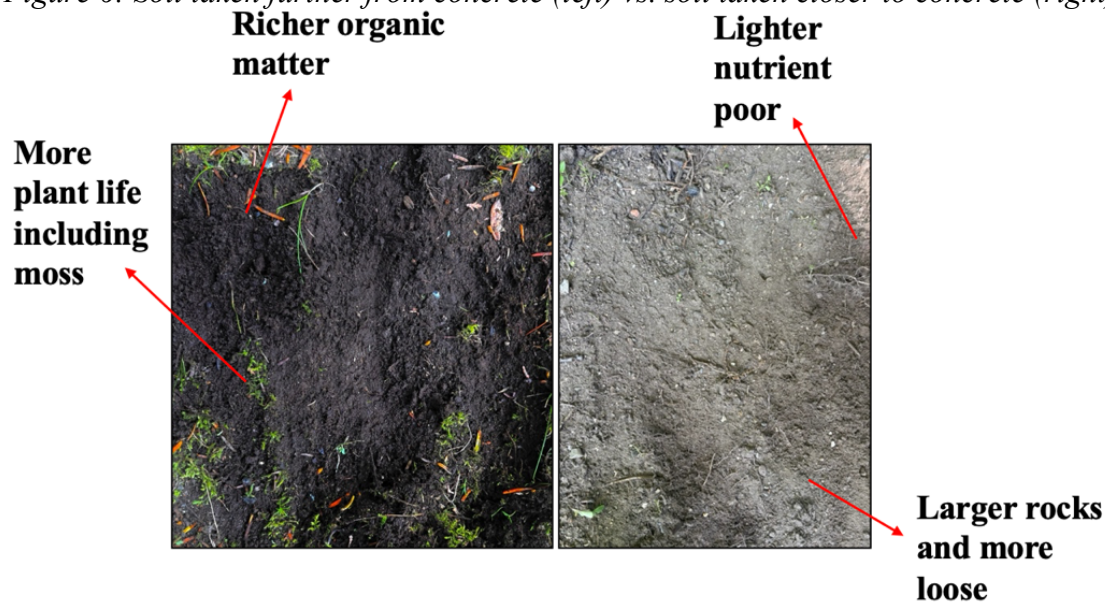
Graph 4: Trendline Plot of Distance from Concrete vs. Nitrogen, Phosphorous and pH



Qualitative observations

Generally, it was noticed that as soil was sampled farther from the source of concrete, it looked richer and darker and was moister.

Figure 6: Soil taken further from concrete (left) vs. soil taken closer to concrete (right)



Criterion E: Analysis and Conclusion

Analysis

Based on the trends, the data collected supports the original hypothesis. Table 1 and Graph 1 both represent the effect of distance from the source of concrete on the pH level. As the distance increases, the pH changes from 10.3 close to the concrete, to 7.1 forty meters away from the source and eventually to 5.6 a hundred meters away. Because concrete contains calcium hydroxide ions which are strongly basic, the pH of the soil close to concrete was high. As the hydroxide ions dissolve in water, the pH rises which then leaches in the soil. However, as the distance from his concrete increased, the leaching of hydroxide ions decreased, so the pH also decreases.

When modeling this chance, a polynomial trendline was chosen as it represents the initial steep drop in pH as the distance increases, but later the pH levels off and plateaus at greater distances. Also, the R^2 was computed giving a value of 0.95 which signifies that the trendline is an accurate model for the data. In addition, standard deviation error bars were calculated ranging from 0.3-0.6. However, at Point X, labeled on Graph 1, the standard deviation was abnormally high, and the datapoint doesn't fit the general pattern of the data, as we would expect a higher value, and thus is an anomaly to the trend.

Also, from graphs 2 and 3 which explore how nutrient levels (Nitrogen and Phosphorous) changed as the distance. Both represent the positive correlation between the distance from concrete and the nutrient levels. Both graphs have a high R^2 value (the R^2 signifies the variance in the data in relation to the trendline), which signifies that the linear trendline associated with the graph fits the data well. Mostly the standard deviation error bars are minimal, however there was great variability between the trials when testing for soil that was 40 meters away and possible represents an anomaly to the trend. Due to the risen pH closer to the concrete, microbial

activity and especially nitrogen-fixing bacteria was inhibited, and thus reduced nitrogen and phosphorus cycling that naturally occurs.

The equipment used had a low uncertainty, which allowed the research question to be accurately assessed. This increased the validity of the experiment, and the systematic sampling strategy further improved the reliability of the findings and prevented bias to be introduced. In other sampling strategies the bias of the individual sampling affects the results, however because the locations of sampling were predetermined there was no personal bias affecting the results.

The summary of the trends is outlined by Graph 4 above (page 13)

Conclusion in Local and Global Context

The reduction of pH and rise of nutrients as distance increases away from a source of concrete has major local implications for Boston. Now that this trend has been identified, this could indicate how soil degrades due urbanization. Thus, soil close to sources of concrete may require remediation or use of fertilizers before planting gardens or trees. Also, schools and community organizations could utilize these findings to promote the awareness of urbanization and how it leads to soil degradation and thus could promote regulations and legislation to improve practices for urban soil conservation (Boston, n.d.). Many plants in the Boston area such as deciduous trees, wouldn't tolerate the high pH when planted near concrete and thus could inform local governmental officials.

Also, globally, these trends could help inform organic and less commercial agriculture practices to avoid the widespread nutrient depletion in soils that could reduce crop yields. As stated in the ESS syllables, this would directly improve global food security as a greater portion of the land is usable. Furthermore, poor soil health can reduce carbon sequestration capacity of soil and further contribute to climate change (Lal, 2004). The data outlined by this study aligns with the UN Sustainable Development Goal 15 which targets soil restoration and sustainable use of land.

Criterion F: Evaluation

Evaluation

The key strength of this investigation is rooted in the systematic nature of the method. The highly precise digital ruler and the compass to ensure that samples were taken exactly 20 meters apart and because 5 soil samples were taken at each distance, the results obtained were very reliable. This allowed the conclusion to be assessed accurately. Also, because the soil was returned to its original location, the ecosystems weren't harmed by this study. However, to further improve, more soil samples could have been collected at each site to enhance the data reliability. Also, samples were only taken at the Chestnut Hill Reservation and to further improve the validity of the results, other samples could be taken in different parks or areas. This limited the scope of my conclusion, as this wasn't generalizable to other locations.

Furthermore, because the soil kits that were used required an estimation of the levels of nitrogen, phosphorus and pH using the color of the solution compared to the color key, there was significant variability in the data. This estimation had a high degree of uncertainty and introduced significant bias in the results. Because this estimation was carried out by me, my overall bias affected how I estimated the concentrations of nutrients. This estimation is an inaccurate method of assessing the concentration of a substance, and thus to improve, this study could have used a digital meter that much more accurately assess the nutrient composition of the soil. This inaccuracy significantly impacted my conclusion, as it was difficult to definitely identify the relationship between distance from concrete and the soil's nutrients. Besides this, the uncertainties on the equipment used was minimal and most likely didn't significantly affect the results of the experiment. Also, the soil samples degrade over time, and thus this could significantly affect the concentrations of different nutrients. This specific variable wasn't accounted for, and thus possible introduced compromised the validity of the experiment, and thus to improve, the method should standardize the time between collection and testing of the soil samples.

Finally, there were significant uncontrolled environmental variables during this experiment. Recent rainfall, use of fertilizers or other pollution sources weren't factored in during the analysis and possibly could have compromised the reliability of the results. This also impacted my conclusion, as these environmental factors may have changed the overall nutrient content of the soil. Thus, to improve this experiment could have possible been conducted in a controlled environmental setting. Furthermore, a grid-based sampling map could have been conducted to ensure even spatial sampling to reduce bias of selecting the appropriate site.

Unresolved Questions and How to Improve This Study

While this pattern was unserved during a snapshot time, it would be interesting to observe the long-term impact of concrete on soil health. Also, only Nitrogen and Phosphorous were tested, and potentially assessing Potassium or Magnesium might further validate these findings. Also, concrete tends to leach other heavy metals, and thus assessing the concentration of heavy metals close to concrete may also be another avenue for research.

Bibliography

Agronomy eUpdate September 8th, 2022 : Issue 923.

(n.d.). https://eupdate.agronomy.ksu.edu/article_new/soil-sample-handling-practices-can-affect-soil-nitrate-test-accuracy-511-4

Andrew, R. M. (2019). Global CO₂ emissions from cement production, 1928–2018. *Earth System Science Data*, 11(4), 1675–1710. <https://doi.org/10.5194/essd-11-1675-2019>

Boston, U. (n.d.). *Environment - UMASS Boston*. <https://www.umb.edu/environment/>

Commonwealth of Massachusetts. (n.d.). *Massachusetts Department of Agricultural Resources*. Mass.gov. <https://www.mass.gov/orgs/massachusetts-department-of-agricultural-resources>

Jburke. (2024, April 29). *The Indigenous History of Boston Harbor - New England Aquarium*. New England Aquarium. <https://www.neaq.org/the-indigenous-history-of-boston-harbor/>

Lal, R. (2004). Soil carbon sequestration to mitigate climate change. *Geoderma*, 123(1–2), 1–22. <https://doi.org/10.1016/j.geoderma.2004.01.032>

Li, S., Yin, J., & Zhang, G. (2020). Experimental investigation on optimization of vegetation performance of porous sea sand concrete mixtures by pH adjustment. *Construction and Building Materials*, 249, 118775. <https://doi.org/10.1016/j.conbuildmat.2020.118775>

Mao, Z., Li, Q., & Wang, Y. (2023). Research on the application of a cement and soil aggregate for the ecological restoration of vegetation in artificial soil. *PeerJ*, 11, e14657. <https://doi.org/10.7717/peerj.14657>

Michelle for Boston. (n.d.). *Climate justice | Michelle Wu for Boston*. <https://www.michelleforboston.com/issues/climate-justice>

National Research Council. (2013a). How much tree canopy does Boston have? In *Urban Forestry: Toward an Ecosystem Services Research Agenda: A Workshop Summary*. The National Academies Press. <https://d279m997dpfwgl.cloudfront.net/wp/2020/01/Boston-2016-Tree-Canopy-Assessment.pdf>

National Research Council. (2013b). How much tree canopy does Boston have? In *Urban Forestry: Toward an Ecosystem Services Research Agenda: A Workshop Summary*. The National Academies Press. <https://d279m997dpfwgl.cloudfront.net/wp/2020/01/Boston-2016-Tree-Canopy-Assessment.pdf>

Palmieri, J. F., Boston Redevelopment Authority, Meade, P., Lima, A., Melnik, M., Avault, J., Blake, N. D., & Borella, N. M. (2011). Boston by the numbers. In *Boston by the Numbers* [Report].

Ritchie, H., Samborska, V., & Roser, M. (2024, February 21). Urbanization. Our World in Data. <https://ourworldindata.org/urbanization>

Rudolph spotted around Boston. (2024, December 9). Boston Preservation Alliance. <https://www.bostonpreservation.org/news-item/rudolph-spotted-around-boston#:~:text=Blue%20Cross/Blue%20Shield%2C%201961,now%20an%20official%20Boston%20Landmark.>

Soil degradation: the silent global crisis. (n.d.). Heinrich Böll Stiftung. <https://eu.boell.org/en/SoilAtlas-soil-degradation>

Soil, S. (2024, June 14). 95% of the Earth's land set to be degraded by 2050 | *Earth.Org*. Earth.Org. <https://earth.org/95-of-the-earths-soil-on-course-to-be-degraded-by-2050/>

United Nations Environment Programme. (n.d.). *Five drivers of the nature crisis*. UNEP. <https://www.unep.org/news-and-stories/story/five-drivers-nature->

- crisis#:~:text=Since%201990%2C%20around%20420%20million%20hectares%20of,lost%20through%20conversion%20to%20other%20land%20uses.
- US Census Bureau. (2021, October 8). *Urban Areas Facts*.
Census.gov. <https://www.census.gov/programs-surveys/geography/guidance/geo-areas/urban-rural/ua-facts.html>
- U.S. Environmental Protection Agency. (2003). *URBAN RUNOFF* [Report]. https://www3.epa.gov/npdes/pubs/nps_urban-facts_final.pdf
- Wang, S., Yuan, X., Li, T., Yang, J., Zhao, L., Yuan, D., Guo, Z., Liu, C., & Duan, C. (2024). Changes in soil microbe-mediated carbon, nitrogen and phosphorus cycling during spontaneous succession in abandoned Pb Zn mining areas. *The Science of the Total Environment*, 920, 171018. <https://doi.org/10.1016/j.scitotenv.2024.171018>
- Xiao, J., Zou, S., Poon, C. S., Sham, M. L., Li, Z., & Shah, S. P. (2025). We use 30 billion tonnes of concrete each year — here’s how to make it sustainable. *Nature*, 638(8052), 888–890. <https://doi.org/10.1038/d41586-025-00568-4>
- Yang, J., & Zhang, G. (2011). Water infiltration in urban soils and its effects on the quantity and quality of runoff. *Journal of Soils and Sediments*, 11(5), 751–761. <https://doi.org/10.1007/s11368-011-0356-1>